

CHAEYOON SHIN

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Last updated: Sep 2026

RESEARCH INTERESTS

- Leveraging diverse biomedical data (multi-omics, clinical registry, patient-generated health data) through computational & statistical modeling
- Quantifying heterogeneity in individual health trajectories and outcomes (e.g., causal machine learning, risk stratification, individualized intervention)
- Mental health & psychiatric phenotypes

EDUCATION

Samsung Advanced Institute for Health Sciences & Technology (SAIHST), Sungkyunkwan University Mar 2026 –

M.S. in Digital Health

Advisor: Dr. Yoonjung Yoonie Joo

Sungkyunkwan University (SKKU) Mar 2022 – Feb 2026

B.E. in Integrative Biotechnology (*Magna Cum Laude*)

The University of Texas at Dallas Aug 2024 – Dec 2024

Fall Exchange Student Program

PUBLICATIONS

Preprints

- [1] **Chaeyoon Shin**, Junghoon Park, et al. Psychological resources confer greatest protection against late-life depression in genetically vulnerable adults. Preprint, June 2026.
<https://doi.org/10.21203/rs.3.rs-9990161/v1>
- [2] **Chaeyoon Shin**, Jinyun Kim, Se In Sung, Yoonjung Yoonie Joo. Heterogeneous Associations of Latency Duration with Neonatal Outcomes After Preterm Prelabor Rupture of Membranes: A Nationwide Cohort Study. Preprint, Sep 2026. <https://doi.org/10.64898/2026.09.07.26361886>

RESEARCH EXPERIENCE

Undergraduate Research Intern Jan 2025 – Feb 2026

Genomics & Neural Network Intelligence for Healthcare (GENNIE) Lab, SKKU

Undergraduate Research Program (URP) Dec 2023 – Aug 2024

Computational Cognitive Affective Neuroscience (COCOAN) Lab, SKKU

- Conducted fMRI data analysis and derived activation maps using MATLAB
- Participated in pilot studies and assisted fMRI experiments

Industry-Academia Collaboration Project Jan – Feb 2024

Lab of Molecular Development and Cell Signaling, SKKU

- Trained in cell culture and experimental techniques in ovarian cancer research

- Performed JC-1 staining, hanging drop assays, and tissue preparation protocols

CONFERENCE PRESENTATIONS

Poster presentation, Korean Society of Medical Informatics (KOSMI) Jun 2026
 “Interplay of Psychological and Genetic Factors in Shaping Late-Life Depression”
 (*co-authored poster was awarded Best Poster)

Poster presentation, American Society of Human Genetics (ASHG) Oct 2026
 “Neuroticism polygenic score marks stronger protective associations between psychological resources and late-life depressive symptoms: evidence from a 35-year sociogenomic cohort”

TEACHING & MENTORING

TA, *Introduction to Healthcare Data Analytics* Spring 2026
 TA, *Introduction to Multiomics Analysis* Fall 2026

AWARDS & HONORS

Scholarships

SKKU Graduate Scholarship (100% tuition support for Master's Program) 2026
 KorUS Educational Exchange Initiative for Youth in STEM (\$9,000/semester) 2024
Issued by Korea Institute for Advancement of Technology (KIAT)
 SKKU Next Generation Scholars Program (100% tuition support for 3rd, 4th undergraduate years) 2024
 SKKU Academic Excellence Scholarship (top 10%) 2023

EXTRACURRICULARS

Member of SKKAI (Sungkyunkwan AI Association), Medical AI Team Sep 2025 – Jan 2026
 • Explainable AI (XAI) research (sparse autoencoders) on pathological foundation models

Member of National Undergraduate Symposium on Biology Aug – Sep 2023
 • Delivered a group presentation on cellular senescence and immunometabolism

SKILLS

Languages: English (TOEFL 107/120, TOEIC 960/990), Korean (Native)
 Programming: R, Python, Linux/Bash